CardioTrack CT-200

Home Blood Pressure Monitor

User & Technical Manual — Version 2.0

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WARNING: Read all instructions carefully before use. For home use only. Not intended as a replacement for professional medical advice.

1. Introduction

The CardioTrack CT-200 is a clinically validated, oscillometric blood pressure monitor designed for home use by adults aged 18 years and older. It measures systolic and diastolic blood pressure, as well as pulse rate, using an upper-arm cuff. The device is intended to assist users in tracking their cardiovascular health over time under the guidance of a healthcare professional.

1.1 Intended Use

The CT-200 is indicated for non-invasive blood pressure measurement in adult patients. It is designed for self-measurement at home and is not intended for use in neonates, children, or patients with arrhythmia unless specifically directed by a physician. Results should be interpreted by a qualified healthcare provider.

1.2 Contraindications

Do not use this device if any of the following apply:

- The patient has a cardiac pacemaker or other implanted electronic device.
- The measurement site has lymphedema, fistula, or intravenous lines.
- The patient is undergoing infusion therapy on the measured arm.
- The cuff is visibly damaged or the tubing is kinked.

1.3 Symbols and Conventions

Throughout this manual, the following conventions are used:

Symbol / Term	Meaning
WARNING	Risk of patient injury if instruction not followed.
CAUTION	Risk of device damage or inaccurate readings.
NOTE	Important information that improves usability.
SYS	Systolic pressure (mmHg)
DIA	Diastolic pressure (mmHg)
PR	Pulse rate (bpm)

2. Device Description

The CT-200 system consists of a main monitor unit, a standard adult upper-arm cuff (fits arm circumferences 22–42 cm), a latex-free inflation bladder, a USB-C charging cable, and a carrying case. An optional large cuff (fits 32–52 cm) is available separately.

2.1 Hardware Components

- Main unit: 130 mm × 90 mm × 50 mm, 280 g (without batteries)
- 3.5-inch colour LCD display (320 × 240 resolution)
- Rechargeable Li-ion battery: 1800 mAh, rated for 350 measurements per charge
- Built-in Bluetooth 5.0 (BLE) for smartphone connectivity
- Internal memory: stores up to 90 readings per user (2 users supported)
- Operating pressure range: 0–300 mmHg
- Accuracy: ±3 mmHg (pressure), ±5% (pulse rate)

2.2 Software Overview

The CT-200 runs firmware version 2.2.0. The firmware manages measurement acquisition, data storage, Bluetooth communication, and safety interlocks. A companion mobile application (CardioTrack App, iOS 14+ / Android 10+) is available for trend tracking and physician sharing.

2.3 Physical Interfaces

The device exposes the following external interfaces:

Interface	Type	Purpose
USB-C Port	Charging only	Battery recharging; data export not supported via USB
Cuff Connector	Proprietary pneumatic	Connects upper-arm cuff to inflation system
BLE Radio	Bluetooth 5.0 LE	Pairing with CardioTrack mobile app
START/STOP Button	Tactile pushbutton	Initiates and aborts measurement cycles
MODE Button	Tactile pushbutton	Cycles through user profiles and history view

3. Safety Requirements

All safety requirements in this section are mandatory and shall be verified during production acceptance testing and post-market surveillance.

3.1 Pressure Safety

The inflation system shall not exceed a maximum cuff pressure of 300 mmHg under any operating condition. The firmware enforces this limit via a hardware pressure relief valve and a software interlock.

If cuff pressure exceeds 290 mmHg during inflation, the firmware shall:

- Immediately open the exhaust valve to initiate deflation.
- Display error code E3 on the LCD.
- Log the event with a timestamp to non-volatile memory.
- Auto-deflate to 0 mmHg within 1.5 seconds of detection.

3.2 Electrical Safety

The CT-200 is classified as a Class II device (IEC 60601-1). The device shall meet IEC 60601-1 Edition 3.2 requirements for basic safety and essential performance. Maximum patient leakage current: 10 µA (normal condition), 50 µA (single-fault condition).

3.3 Electromagnetic Compatibility

The CT-200 shall comply with IEC 60601-1-2 for EMC. The device must maintain essential performance in environments with RF field strengths up to 3 V/m (80 MHz – 2.5 GHz). The BLE radio shall not interfere with other medical devices within 1 metre.

3.3 Battery and Charging Safety

The integrated Li-ion battery shall be protected against overcharge, over-discharge, and short-circuit conditions by the battery management IC. Charging shall cease automatically when the battery reaches 100% state of charge. The device shall display a low-battery warning when charge falls below 15%.

3.4 Alarm Conditions

The device shall generate a visual alarm (flashing display) for the following conditions:

- E1: Cuff not detected or improperly connected
- E2: Arm movement detected during measurement (excessive motion artifact)
- E3: Cuff pressure exceeded safe limit — threshold 290 mmHg (see 3.1)
- E4: Battery critically low (below 5%)
- E5: Internal sensor fault
- E6: Bluetooth pairing failure after 3 consecutive attempts
- E7: Measurement timeout — cycle not completed within 120 seconds

4. Measurement Procedure

Follow these steps to ensure accurate and reproducible blood pressure measurements.

4.1 Patient Preparation

- Rest quietly for at least 5 minutes before measurement.
- Avoid caffeine, smoking, and exercise for 30 minutes prior.
- Sit upright in a chair with back support; feet flat on floor.
- Support the arm at heart level on a flat surface.
- Remove tight clothing from the upper arm.

4.2 Cuff Application

Place the cuff on the bare upper arm 2–3 cm above the antecubital fossa (elbow crease). The artery marker on the cuff must align with the brachial artery. Tighten the cuff so that two fingers can be inserted between the cuff and the arm.

4.3 Measurement Acquisition

Press START/STOP. The device will inflate the cuff automatically to the target pressure, then gradually deflate while the oscillometric algorithm detects oscillations corresponding to systolic and diastolic pressures. The measurement cycle takes approximately 30–60 seconds. Do not move or speak during measurement.

4.4 Reading Results

Upon completion, the display shows SYS / DIA (mmHg) and PR (bpm). Values are automatically stored in the current user's memory slot with a date/time stamp. An IHB (Irregular HeartBeat) indicator will appear if the pulse rhythm was inconsistent during measurement.

5. Alarm and Error Handling

This section describes the complete set of alarms, their triggers, and required device responses. All alarms shall be tested during acceptance testing.

5.1 Error Code Reference

Error Code	Trigger Condition	Device Response	User Action
E1	Cuff not detected	Halt inflation, display E1	Re-attach cuff
E2	Motion artifact > threshold	Abort measurement, display E2	Repeat measurement
E3	Pressure ≥ 290 mmHg	Deflate within 1.5 s, display E3, log event	Contact service
E4	Battery < 5%	Display E4, allow one more measurement	Charge device
E5	Sensor fault	Halt, display E5, log fault code	Power cycle; if persists, service
E6	BLE pairing fail (3×)	Display E6, disable BLE for 60 s	Retry pairing
E7	Measurement timeout (120 s)	Abort, deflate, display E7	Retry; check cuff seal

5.2 Watchdog Timer

The firmware implements a hardware watchdog timer with a 4-second timeout. If the main application loop fails to service the watchdog within this period (e.g., due to a software hang), the watchdog resets the microcontroller and logs an unclean shutdown event.

5.3 Measurement Timeout

If a measurement is not completed within 120 seconds of initiation (e.g., cuff leakage prevents the target pressure from being reached), the firmware shall abort the cycle, display error code E7, and fully deflate the cuff.

6. Data Management

The CT-200 stores measurement data locally and optionally synchronises it to the CardioTrack mobile application over BLE.

6.1 Local Storage

Internal memory holds up to 90 readings per user profile. When memory is full, the oldest reading is overwritten (circular buffer). Each record contains: date, time, SYS, DIA, PR, IHB flag, and a measurement quality score (0–100).

6.2 Bluetooth Synchronisation

After a successful measurement, the device advertises a BLE GATT notification containing the latest reading. The CardioTrack app, if in range and paired, receives the notification and stores the reading in the app's local database. Full history sync can be triggered manually from the app.

6.3 Data Privacy

All BLE communications are encrypted using AES-128 CCM as per the BLE specification. The device does not collect or transmit personally identifiable information (PII). User profile names stored on the device are limited to 8 alphanumeric characters.

7. Maintenance and Service

7.1 Cleaning

Clean the exterior of the monitor unit with a soft cloth lightly dampened with 70% isopropyl alcohol. Do not immerse the device in liquid. The cuff may be hand-washed in lukewarm water and air-dried. Do not machine-wash the cuff.

7.2 Calibration

The CT-200 requires calibration every 2 years or after any repair that affects the pressure transducer. Calibration must be performed by an authorized service centre using traceable reference equipment. Field calibration by the end user is not permitted.

7.3 Disposal

Dispose of the CT-200 in accordance with local electronic waste (WEEE) regulations. The Li-ion battery must be removed before disposal and recycled separately. Do not dispose of batteries in municipal waste.

8. Technical Specifications

Parameter	Specification
Measurement Method	Oscillometric
Measurement Range (Pressure)	20–285 mmHg
Measurement Range (Pulse)	40–200 bpm
Accuracy (Pressure)	±3 mmHg
Accuracy (Pulse Rate)	±5% of reading
Max Safe Cuff Pressure	300 mmHg (hardware limit), 290 mmHg (software limit)
Cuff Size (Standard)	22–42 cm arm circumference
Display	3.5-inch colour LCD, 320×240
Battery	Li-ion 1800 mAh, USB-C charging
Battery Life	≥350 measurements per charge
Memory	90 readings × 2 users
Connectivity	Bluetooth 5.0 LE
Operating Temperature	10–40 °C
Storage Temperature	-20–60 °C
Ingress Protection	IP20
Dimensions	130 × 90 × 50 mm
Weight	280 g (without batteries/cuff)
Regulatory	IEC 60601-1 Ed. 3.2, IEC 60601-1-2, ISO 81060-2

9. Regulatory Compliance

The CT-200 has been designed and manufactured in accordance with the following standards and regulations. Compliance documentation is available upon request.

- IEC 60601-1:2005+AMD1:2012+AMD2:2020 (General requirements for basic safety and essential performance)
- IEC 60601-1-2:2014+AMD1:2020 (Electromagnetic disturbances)
- ISO 81060-2:2018/AMD1:2020 (Non-invasive sphygmomanometers – clinical investigation)
- EU MDR 2017/745 (European Medical Device Regulation)
- FDA 21 CFR Part 820 (Quality System Regulation)
- ANSI/AAMI SP10:2002/(R)2008 & A1:2003/(R)2008 & A2:2006/(R)2008

9.1 Declaration of Conformity

Tri9T Medical Devices Ltd. declares that the CT-200 conforms to the essential requirements of Annex I of the EU MDR 2017/745. The declaration of conformity is available at https://tri9t.example.com/doc/CT200.

9.2 Authorized Representative

EU Authorised Representative: MedRep EU GmbH, Berliner Str. 42, 10115 Berlin, Germany. UK Responsible Person: MedRep UK Ltd., 10 Harley Street, London, W1G 9PF.

10. Revision History

Revision	Date	Author	Description
A	2025-01-10	J. Smith	Initial release
A.1	2025-03-15	L. Patel	Corrected cuff size table, added E6 error code
B	2025-09-01	R. Nguyen	Updated pressure safety threshold to 290 mmHg, tightened deflat

10.1 Firmware Update Procedure

Firmware updates are distributed via the CardioTrack mobile application. To update firmware:

- Ensure device battery is above 50% before initiating an update.
- Open the CardioTrack app and navigate to Settings > Device > Firmware Update.
- The app will download the update package and transfer it to the device over BLE.
- The device will verify the package signature before applying the update.
- Do not power off the device or close the app during the update process.
- After a successful update, the device will restart automatically.

WARNING: Interrupting a firmware update may corrupt the device firmware and require factory servicing. Always ensure stable BLE connectivity before updating.